

INTERPOL Guidelines for Digital Forensics Labs (DFL)

Section 3 - Infrastructure & Security

1. Have an uninterruptible power supply (UPS) for critical systems. (3.2.1)
2. Maintain physical security in the lab, ensuring restricted access. Allowing only authorised personnel to access the lab. (Biometric locks, camera monitoring etc.) (3.2.2)
3. Ensure the lab meets environmental standards (protection from dust, humidity, and static). (3.2.2)
4. Secure evidence storage room or safe box - To safely store electronic evidence. Access restricted to certain DFL staff only. (3.2.4)
5. Protect forensic item from unauthorized network access – Where isolated network is not available (3.2.4)
6. Maintain an incident log for all activities within the lab. (Evidence delivery, investigations undertaken, times, dates etc.) (3.4.1)
7. Tools and accessories to help with performing cases and evidence storage. (Tamper-proof stickers, faraday bag, power extensions, camera etc.) (3.4.3)

Section 5 - Lab Procedures

1. Use forensic acquisition hardware (write blockers, forensic duplicators). (5.1.2.2)
2. Follow recommendations of imaging tool functionality and the imaging Format. (5.1.2.3)
3. Document all actions and data extracted from devices. (5.1.3.4)
4. Examinations should be performed on copy of data instead of original where possible (5.2.1)
5. Automate some of the investigation steps where possible (Data exfiltration, finding deleted files etc.) (5.2.3.3)
6. Follow procedures to find data from different traces on the system. (5.3.1.2)
7. Visualisation of events where needed (could help in an investigation process.) (5.3.1.5)
8. Findings recording ensuring they can be recreated and validated. (5.4.1)

Section 6 – Quality Assurance

1. Ensure components checklist is met (Laboratory, Facility & Environment, Equipment, Staff, Forensic Method, Service Request, Evidence Handling, Forensic Result) (6.1)
2. DFL Accreditation (Voluntary and not mandatory - ISO/IEC 17025) (6.2)

Appendix K – Electronic Evidence

1. Labelling – Ensure all evidence is labelled properly and uniquely once it reaches the DFL. (Appendix K - 1)

2. Sealing – Must be sealed using a proper container which can detect any access to the evidence ([Appendix K - 2](#))
3. Document – Full record of electronic evidence must be kept, including inventory list in the DFL, chain of custody records etc. ([Appendix K - 3](#))
4. Not Left Unattended – Evidence not currently being investigated must be stored away securely. ([Appendix K - 4](#))
5. Keep away from source of contamination – Evidence must be properly taken care of away from things like water, heat etc. ([Appendix K - 5](#))
6. Storage – Evidence must be securely stored with limited access. ([Appendix K - 6](#))
7. Preservation – Copy of original data should be kept to preserve integrity, hash values must be same for copy and original. ([Appendix K - 7](#))

(INTERPOL, 2019) (ISO/IEC, 2017)

Interpol Guidelines	Physical Lab			Cloud Lab		
	Meets	Could Meet	Can't Meet	Meets	Could Meet	Can't Meet
Section 3. Infrastructure & Security						
1. UPS for critical systems		X				X
2. Physical Security		X				X
3. Environmental Controls		X				X
4. Secure Evidence Storage		X				X
5. Protected Network Access	X			X		
6. Incident Log	X			X		
7. Tools & Accessories		X				X
Section 5. Lab Procedures						
1. Forensic Acquisition Hardware		X			X	
2. Imaging Recommendations		X			X	
3. Documentation of all actions	X			X		
4. Perform analysis on data copy	X			X		
5. Automation of some processes	X			X		
6. Follow procedures to find hidden data		X			X	
7. Visualisation of Events	X			X		
8. Recording of findings for recreation	X			X		
Section 6 – Quality Assurance						
1. Components Checklist		X			X	
2. DFL Accreditation		X			X	
Appendix K – Electronic Evidence						
1. Labelling	X			X		
2. Sealing		X				X
3. Document	X			X		
4. Not Left Unattended	X			X		
5. Keep from Contamination		X		X		
6. Storage	X			X		
7. Preservation	X			X		

Interpol Guidelines	Physical Lab			Cloud Lab		
	Meets	Can Meet	Can't Meet	Meets	Can Meet	Can't Meet
Infrastructure & Security	70%	30%		30%		70%
Lab Procedures	63%	37%		63%	37%	
Quality Assurance		100%			100%	
Electronic Evidence	71%	29%		86%		14%

References

INTERPOL, 2019. *Interpol*. [Online]

Available at:

https://www.interpol.int/content/download/13501/file/INTERPOL_DFL_GlobalGuidelinesDigitalForensics

[Accessed 2025].

ISO/IEC, 2017. *General requirements for the competence of testing and calibration laboratories*.

[Online]

Available at: <https://www.iasonline.org/wp-content/uploads/2021/02/ISO-IEC-17025-2017-IAS.pdf>